PHYSICS I / STUDENT DIAGNOSTIC

Newton's Laws - Check Your Reasoning

PlusOne demo quiz | 8 multiple choice + 2 reasoning questions | 20-25 min

Name / alias: _____ **Date:** _____

Choose one answer for questions 1-8. Show reasoning for questions 9-10. This is a **formative diagnostic**, not an exam grade. Work individually before the group discussion.

Assume: an approximately inertial frame, SI units, and g = 9.81 m/s² where needed.

Q01 A puck slides right at 2.0 m/s on an ideal frictionless horizontal surface. After it is released, the net horizontal force on it is zero. What happens next?

A. It gradually slows to rest.

B. It continues right at 2.0 m/s.

C. It accelerates toward the right.

D. It speeds up, then slows down.

Q02 A 2.0 kg cart is pulled with 10 N to the right while an opposing 4 N force acts to the left. What is its horizontal acceleration?

A. 7.0 m/s² right

B. 5.0 m/s² right

C. 3.0 m/s² right

D. 3.0 m/s² left

Q03 A 2.0 kg cart and a 4.0 kg cart each experience the same net horizontal force. Compared with the 4.0 kg cart, the acceleration of the 2.0 kg cart is...

A. the same.

B. twice as large.

C. half as large.

D. four times as large.

Q04 A book rests on a table. Which pair is a Newton's third-law force pair?

A. Earth pulls the book down; the table pushes the book up.

B. Earth pulls the book down; the book pulls Earth up.

C. The table pushes the book up; the book has zero acceleration.

D. The book pushes the table down; Earth pulls the book down.

Q05 An elevator carries a person upward at a constant speed. Neglect air resistance. What is the scale's upward normal force N on the person?

A. N = 0

B. N < mg

C. N = mg

D. N > mg

Q06 You push a box horizontally to the right with 12 N. It remains at rest on a rough horizontal floor, and no other horizontal forces act. What is the horizontal static friction force on the box?

A. 0 N, because the box does not move.

B. 12 N to the left.

C. The maximum possible static friction, regardless of the push.

D. 12 N to the right.

Q07 During a head-on collision, a heavy truck exerts a force on a small car. How does the force exerted by the car on the truck compare at the same instant?

A. Same magnitude, opposite direction, on the other body.

B. Smaller magnitude because the car is lighter.

C. Larger magnitude because the car changes speed more.

D. Both forces act on the car and cancel.

Q08 A block is released from rest on a frictionless ramp inclined at 30° to the horizontal. Use g = 9.81 m/s². What is its acceleration along the ramp?

A. 0 m/s²

B. 8.50 m/s² down the ramp

C. 4.91 m/s² down the ramp

D. 9.81 m/s² down the ramp

Q09 A classmate says: "If a cart keeps moving, there must be a force pushing it forward." Explain whether this is correct for a cart moving at constant velocity on an ideal frictionless track. Then explain why everyday carts on rough floors often stop without being pushed.

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Q10 Two blocks of masses 2.0 kg (A) and 3.0 kg (B) touch on a frictionless horizontal surface. A horizontal 10 N force pushes block A to the right. Find (i) the acceleration of the two-block system, (ii) the contact force of A on B, and (iii) the contact force of B on A. Name which two forces form a third-law pair.

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Peer-learning discussion cards

Discuss only after the individual diagnostic. Every participant should make at least one reasoned contribution. The AI facilitator should help you question each other instead of handing you the completed answer.

Motion without a push

Starting disagreement: Student A says, "If the puck is still moving, something must keep pushing it." Student B says, "No, because acceleration and velocity are different."

Group task: Discuss what would happen on an ideal frictionless surface and on a rough floor. Each student must use the phrase "net force" correctly.

Truck and car disagreement

Starting disagreement: Student A says, "The truck exerts more force because it is heavier." Student B says, "The forces are equal but accelerations can differ."

Group task: Label the two interacting bodies and identify the forces for each. Explain which mass matters for acceleration.

The static-friction trap

Starting disagreement: Student A says, "There is no friction because the box does not move." Student B says, "The friction always equals mu_s times the normal force."

Group task: A 12 N horizontal push does not move the box. Use Newton's second law to decide the actual friction force and explain why both statements can be wrong.

Independent follow-up

After the discussion, solve a related but different question alone. This checks whether you can transfer the idea, not just repeat a group answer.